

Response to Technical Review dated 29 July 2026

GFNO-PINO: A Topology-Disjoint Benchmark Isolating Physics Fine-Tuning, Topology Conditioning, and Spectral Filtering in AC-OPF Graph Surrogates

We thank the reviewer for the precise formulation, architecture, and reproducibility audit. We corrected the manuscript and regenerated its tables, figures, PDF, integrity manifest, and submission archive. We did not invent a repository DOI or add unexecuted topology experiments. The resulting scope is more conservative and matches the archived code and evidence.

1. **The N-1 connectivity equation omitted fixed transformers.**

Response. Equation (3) now tests connectivity on the active full branch set $\{a \in \mathcal{B} : s_a(z) = 1\}$, with $\mathcal{B} = \mathcal{L} \cup \mathcal{X}$. This matches the data generator, which retains transformer and parallel-circuit edges while screening line outages.

2. **The neural-operator/FNO terminology exceeded the evidence.**

Response. We now call the method a topology-conditioned, physics-informed Chebyshev graph surrogate. We state that its localized polynomial layer is structurally ChebNet, that each grid size has a separate checkpoint, and that no mesh-refinement, discretization-transfer, or shared mixed-size model was tested. “GFNO-PINO” is retained only as the archived experiment identifier. The abstract, contributions, methodology, limitations, conclusion, model docstrings, and README were revised consistently.

3. **Branch angle limits were incorrectly described as predictor inputs.**

Response. The formulation now separates predictor metadata $\mathcal{M}_{\text{pred}}$ from physics/evaluation metadata $\mathcal{M}_{\text{phys}}$. The former contains box limits, incidence, economics, generator identity, and slack information. Branch angle limits and exact Y, Y_f, Y_t belong only to the latter. This matches the implemented ten-channel edge tensor exactly.

4. **Logistic heads cannot attain active endpoints.**

Response. The hard-layer section now states that finite logits map to the open interval and therefore cannot reproduce an endpoint exactly. We evaluated all 20 GFNO-PINO checkpoints using a 10^{-4} p.u. near-bound definition. A new table reports reference active fractions and active/inactive MAEs for V, P^g, Q^g in every case. The executable `analyze_active_bounds.py` and its CSV are included. We discuss projection/hard-sigmoid heads as a retraining-dependent alternative rather than changing the archived experiment retrospectively.

5. **Baseline training and checkpoint selection were underspecified.**

Response. A new protocol table gives fitting rows/topologies, validation rows/topologies, training objective, checkpoint score, early-stopping status, and exact parameter counts for every model and case. It discloses that the fixed PINN uses a seeded 10/2 split of base-only rows, all other models use the topology-held-out validation ID, the data-only model is trained only on labels but selected with the common physics-aware validation score, and all 30 epochs run without early stopping.

6. **Topology selection and scenario generation were not reproducible enough.**

Response. The manuscript now states the exact pre-split rule: retain the base case plus the first 11 non-islanding outages in pandapower line-index order, then shuffle outage IDs with NumPy `default_rng(2026)` and floor the 0.70/0.15/0.15 fractions. No random or severity-based selection preceded this cap. We explicitly limit representativeness because the selected

set is not severity-stratified and no descriptors were produced for excluded outages. The graph-kernel length scales and the full synthetic diurnal construction (clear-sky sinusoid, Beta cloud factor, Gaussian site factor, clipping) are also specified.

7. Training and DC-OPF settings were incomplete.

Response. The revised setup reports AdamW learning rates, weight decay, gradient clipping, the 30-epoch cosine schedule, adaptive-weight Adam settings, dropout, edge-conditioner dimensions, and the absence of warm restarts and early stopping. DC-OPF is identified as pandapower 3.5.4 `rundcopp`, default PYPOWER backend, and 10^{-10} power tolerance, with its active-power/angle model and omitted voltage/reactive variables stated explicitly.

8. Five-seed bootstrap intervals were overinterpreted.

Response. They are renamed descriptive seed-resampling intervals and are not called confidence intervals. The revised table prints all five paired effects for every case. The text reports that all 20 observed effects are positive while explicitly declining population-level coverage or robustness claims. The adaptive-weight figure now uses distinct line styles and markers so the economic curve remains identifiable.

9. Latency and data-availability wording required correction.

Response. “Times lower” is replaced by an $8.3\text{--}138.1\times$ raw device-specific latency ratio (speedup), with the existing hardware and omitted-verification caveats retained. The availability section names the exact ZIP and companion checksum JSON and states honestly that a permanent repository DOI has not yet been assigned; it must be added before publication.

The verification suite passes all 25 tests after these changes.